

ARE YOUR COMPUTERS WELL PROTECTED?

Failure of the information network may severely paralyse a company's working. You can prevent damage to your computer network due to lightning by following some simple recommendations given in this article

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Lightning is an unpredictable natural phenomenon. It is common in hilly areas, coastal areas and high-rise towers. The effects of lightning may vary from place to place and user to user. External lightning protection only protects against fire and damage to the building. But it cannot prevent damage to electronic equipment due to high voltages induced into the house/office wiring via electric mains.

Similarly, there is a common threat of high-voltage surge as a result of an HT overhead cable falling on an LT network, or a sudden surge in input supply beyond acceptance bands of the UPS systems.

Another typical cause for dysfunction of electronic devices and information technology systems is switching operations in the power supply network. Several kilovolts of energy may get coupled to your electric mains supply due to switching on/off of heavy inductive or capacitive loads.

Remember that the most aggravating danger concerns not only households but also companies which are dependent on fully functioning technical systems. Powerful information networks are the backbone of our

modern industrial society. Failure of such systems may severely paralyse a company's working. You can prevent damage to your computer network by following the recommendations given below.

Look hard at your UPS specifications

The UPS industry has gone through a major transformation. There are new and better conversion technologies available. The users, however, are not fully aware of foolproof safety requirements for computer networks. There are safety issues on which UPS vendors sometimes remain silent. The result is that computer networks connected with the given UPS remain unprotected.

Recently, one of our clients, a reputed hotel in Udaipur, suffered major network damage due to lightning. They were using a 6kVA UPS manufactured by a reputed company. Interestingly, when lightning struck, the UPS was switched off. Despite that, each and every computer and other devices connected to the given UPS burnt out. The UPS also got damaged.

After getting answers to a few of my initial queries, I realised that the customer had not given due importance to the various safety issues and

had presumed that those safety issues were built into the UPS. And this very customer is not alone. There are so many users who buy UPS systems off the shelf unaware of finer safety issues or driven by lower cost.

Let us look at this case. While the UPS was 'off,' computer networks were still connected with the UPS output point. During lightning, the neutral link received several kilovolts as induced voltage.

The UPS output was not isolated from the incoming line input supply as the UPS did not have a built-in isolation transformer. Due to which, there was no path for the induced energy to return and eventually damage occurred. Among the various recommendations I gave to this client are:

1. Install super-isolation transformer at the input of the UPS
2. Install Class B and Class C surge arrestors at the input
3. Install exclusive array of lightning arrestors on the building top
4. Provide for exclusive pair of dedicated earth for computer network
5. Use double-pole isolation when switching off the UPS

Look at Figs 1 and 2.

In Fig. 1, there is no isolation transformer between input/UPS and UPS output/load. In such a case, the incoming supply is rectified, boosted and converted into sine wave again and fed to the load. Incoming neutral is the only return path for the various return currents. This scheme does not provide return paths to various noise filters used in the computers. There is no possibility of a computer network

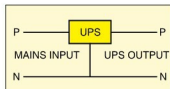


Fig. 1: UPS with non-isolated output

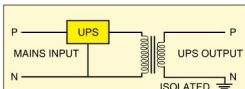


Fig. 2: UPS with isolated output